ThreadEye AI

কাপড় পরীক্ষার স্মার্ট সহকারী — receiving inspection for garment factories.

Problem: find and document fabric faults before they become rejected garments or disputed mill claims.

Journey: roll registration camera/video detector inspector review 4-point verdict photo report claim pack approved cutting instructions supplier scorecard.

Architecture: Next.js + FastAPI + PostgreSQL. YOLOX-S ONNX Runtime CPU. Docker edge profile; a separate Vercel/Supabase simulated cloud demonstration. Bangla and English. Append-only audit chain.

Measured laboratory evidence: current line-scan regression roll: 4 of 7 faults found, 0 false alarms. One crease and two broken yarns are missed. Video is simulated. The regression roll has been reused in development; these results do not establish factory-wide accuracy or business savings.

Human control: inspector confirms/rejects/adds faults; supervisor makes commercial and cutting decisions; cutting master acknowledges instructions. Daily reference checks and benchmark history expose failures.

Impact validation: paired manual/assisted pilot protocol and calculator are available. Time saved, usability gain, yield and financial savings are not claimed until measured.

Demonstration: https://threadeye-web.vercel.app (confirm deployed revision with the team). Owner: Shashwata Nandi and the ThreadEye AI team.

SIMULATED